

Dynamic oxygen mapping in the root zone by fluorescence dye imaging combined with neutron radiography

Nicole Rudolph · Hanna G. Esser · Andrea Carminati ·
Ahmad B. Moradi · André Hilger · Nikolay Kardjilov ·
Stefan Nagl · Sascha E. Oswald

Received: 5 May 2011 / Accepted: 21 July 2011 / Published online: 13 September 2011
© Springer-Verlag 2011

Abstract

Purpose The rooted zone of a soil, more precisely the rhizosphere, is a very dynamic system. Some of the key processes are water uptake and root respiration. We have developed a novel method for measuring the real-time distribution of water and oxygen concentration in the rhizosphere as a biogeochemical interface in soil. This enables understanding where and when roots are active in respect to root respiration and water uptake and how the soil responds to it.

Responsible editor: Michael Kersten

N. Rudolph (✉) · H. G. Esser · S. E. Oswald
Institute of Earth and Environmental Science,
University of Potsdam,
Karl-Liebknecht-Str. 24-25,
14476 Potsdam, Germany
e-mail: nrudolph@uni-potsdam.de

A. Carminati
Department Hydrogeology, Helmholtz Centre
for Environmental Research—UFZ,
Permoser Str. 15,
04318 Leipzig, Germany

A. B. Moradi
Department of Land, Air and Water Resources,
University of California,
1 Shields Avenue,
Davis, CA 95616, USA

A. Hilger · N. Kardjilov
Helmholtz Centre Berlin—HZB,
Hahn-Meitner-Platz 1,
14109 Berlin, Germany

S. Nagl
Institute of Analytical Chemistry, University of Leipzig,
Johannisallee 29,
04103 Leipzig, Germany

Materials and methods We used glass containers ($15 \times 15 \times 1$ cm), which were filled with a quartz sand mixture. Sensor foils for fluorescence dye imaging of O_2 were installed on the inner side of the containers. A lupine plant was grown in each container for 2 weeks under controlled conditions. Then we took time series of fluorescence images for time-lapsed visualization of oxygen depletion caused by root respiration. Changing water content was mapped in parallel by non-invasive neutron radiography, which yields water content distributions in high spatial resolution. Also it can detect the root system of the lupine plants. By this combined imaging of the samples, a range of water contents and different oxygen concentration levels, both induced by root activities, could be assessed.

Results and discussion We monitored the dynamics of these vital parameters induced by roots during a period of several hours. We observed that for high water saturation, the oxygen concentration decreased in parts of the container. The accompanying neutron radiographies gave us the information that these locations are spatially correlated to roots. Therefore, it can be concluded that the observed oxygen deficits close to the roots result from root respiration and show up while re-aeration from atmosphere by gas phase transport is restricted by the high water saturation.

Conclusions Our coupled imaging setup was able to monitor the spatial distribution and temporal dynamics of oxygen and water content in a night and day cycle. This reflects complex plant activities such as photosynthesis, transpiration, and metabolic activities impacting the root–soil interface. Our experimental setup provides the possibility to non-invasively visualize these parameters with high resolution. The particular oxygen imaging method as well as the combination with simultaneously mapping the water content by neutron radiography is a novelty.

Keywords Fluorescence imaging · Neutron radiography · Oxygen mapping · Rhizosphere · Root respiration · Water distribution

1 Introduction

In soils and sediments, there is a strong coupling between biogeochemical processes and local, often heterogeneous, distribution of water, electron acceptors, acids, nutrients, and pollutants. Both sides are closely related and affect each other from the microscopic to the field scale. Oxygen is one of the most important factors in biological systems. Oxygen distribution is highly variable in soils, with concentrations ranging from aerobic conditions down to strictly anaerobic conditions (Hinsinger et al. 2009). Knowledge of oxygen concentration gradients is of prior importance in understanding the function and regulation of microbial communities (Klimant et al. 1995) and plants.

Oxygen undergoes a net transfer from atmosphere to soil by advective and diffusive processes. Under dry soil conditions oxygen flux mainly occurs via gas phase transport. In contrast, in the saturated capillary fringe and groundwater, the transport occurs in the aqueous phase and thus is substantially slower than for drier conditions. Soil is a complex three-phase system with varying degrees of spatial and temporal heterogeneity of physical and chemical properties (Luster et al. 2009). Typical soil structures such as aggregates, clay, layers, macropores, and wettability differences occurring in natural soils imply that heterogeneity and patchiness is a general feature (Brune et al. 2000). Plant roots are an additional source of heterogeneity in soils. Roots are locations of high biogeochemical activity and strongly influence the surrounding soil. This includes exudation of organic acids, respiration activity, and uptake of water (Hinsinger et al. 2009). Root-mediated processes often show a strong dependency from each other. For example, root water uptake may decrease when roots are exposed to anaerobic conditions (Tournaire-Roux et al. 2003).

Blossfeld and Gansert (2007) pointed out the need to avoid any disturbance of the natural conditions of the biogeochemical micropatterns. Due to the destructive character of conventional methods, it is difficult to assess the spatial distribution and temporal dynamics of the variables of interest. By applying non-destructive measurements, it is possible for particular parameters to overcome these limitations. Common example is X-ray CT, mostly suited to map the distribution of material density. Other imaging methods have recently been introduced for soil investigations, and two of them are fluorescence imaging and neutron radiography.

In the last two decades, optical sensors for chemical species found increasing use for plant (Blossfeld and

Gansert 2007), marine (Glud et al. 1996; Precht et al. 2004), and microbiological research (Glud et al. 1998). The receptor part of optical chemical sensors is based on chemical species that show a change in their optical properties (e.g., absorption or emission intensity at particular wavelengths) depending on local environment conditions. The sensitivity and selectivity toward a specific analyte is also influenced by the polymeric matrix in which the sensor molecule is embedded. For oxygen, this mechanism is the basis for a point measurement by laser excitation and lifetime measurement at optode spots, which is increasingly applied in practice (Blossfeld and Gansert 2007). Optical sensors have been developed for variety of important environmental analytes such as pH, CO₂, temperature, O₂, and various ionic species (Fuller et al. 2003).

Fluorescence-based chemical sensors have many advantages compared to other methods. They show a fast response; they are highly sensitive, reversible, and are capable of spatially resolved data acquisition (“imaging”) (Baleizão et al. 2008). Their drawbacks include limited photostability of the indicator dyes (Schroeder et al. 2007). Inaccuracies may also arise from camera drift (Borisov et al. 2006) or non-homogeneous lighting (Liebsch et al. 2000). Applications of fluorescence sensors are numerous. In aquatic systems, for example, Glud et al. (1998) studied the O₂ dynamics at the base of a biofilm. At the water–sediment interface in marine systems, Precht et al. (2004) looked at oxygen distribution in permeable sediments forced by advective pore water exchange, and Schroeder et al. (2007) developed a dual sensor for imaging dynamics of pO₂ and pH. The latter was also mapped for smaller parts of a root system (Blossfeld and Gansert 2007; Schreiber 2010). Moreover, optical sensors have been used to study the development of oxic zones for aquatic plants (Jensen et al. 2005) and very recently Blossfeld et al. (2011). In contrast to wetland plants, which are extensively studied, little has been documented for changes in partly saturated systems.

Neutron imaging has a long history with more than seven decades of research. Neutrons are very efficiently attenuated by materials rich in hydrogen. This makes neutron radiography an optimal tool to investigate water and root distribution in soils. The considerable progress in recent times is mainly due to the developments in digital image recording (Strobl et al. 2009). CCD camera-based detector systems have several advantages, like low acquisition times, good reproducibility, and a high dynamic range (Lehmann et al. 1999). Applications range from non-destructive testing of industrial components to scientific investigations in biology, geology, material science, and physics (Kardjilov et al. 2005). Recently, several investigators showed the applicability of neutron radiography in soil science with special focus on water- and plant-related research. Deinert et al. (2004) used

neutron radiography to study variations of moisture content across a wetting front moving through porous media. Matsushima et al. (2009) showed the applicability of neutron radiography to monitor the velocity of water transport inside rose plants. Oswald et al. (2008) reported that neutron radiography (NR) is a promising imaging technique for studying water relations between soils and plants. Because of its high sensitivity to water, tomography by neutrons can provide accurate images of water movement induced by root water uptake (Moradi et al. 2011; Tumlinson et al. 2008).

In this study, we have developed a non-invasive fluorescence imaging technique to quantify oxygen content distribution and its temporal changes in situ with high spatial resolution. Moreover, we imaged simultaneously the water content distributions by neutron radiography to visualize and quantify the inter-related pattern of oxygen consumption and water distribution in complex aerobic root–soil systems. This combination allowed also for a direct interpretation, if gas phase re-supply of oxygen was restricted or not. Our aim was to capture the in situ oxygen and water distributions with a spatial and temporal resolution appropriate for the rhizosphere. We successfully mapped the effects of root water uptake and respiration under restricted aeration in sand packings. This oxygen mapping by fluorescence dye imaging is designed to visualize also oxygen distributions at other biogeochemical interfaces and is suited to work in natural soils also.

2 Materials and methods

2.1 Fluorescence oxygen imaging

Oxygen-sensitive dyes are based on the ability of oxygen to act as fluorescence quencher, thereby decreasing the fluorescence quantum yield (Carraway et al. 1991a). The optical properties depend on the choice of fluorescent dyes and its carrier polymer. For example, metalloporphyrins dyes with fluorosubstituents typically show increased photostability and slow photodecomposition due to their electron-withdrawing properties that prevent oxidation in the excited state (Schroeder et al. 2007). Suitable fluorescence dyes possess high chemical stability (Carraway et al. 1991b), high quantum yields (Amao and Okura 2000), fast response, and recovery times (Mills and Lepre 1997). Among the most viable sensors that have been used in literature are ruthenium (II) diimine (Liebsch et al. 1999), fullerene (Nagl et al. 2007), platinum(II) as well as palladium(II) porphyrin complexes (Amao and Okura 2000; Borisov et al. 2006). The optical sensor response is monitored by changes in fluorescence intensity, lifetime, or spectral distributions (Fuller et al. 2003). These dye complexes have favorable

properties. They provide large Stokes shifts, which means the absorption and emission signals can be separated spectrally (Borisov et al. 2006) and absorption/emission bands in the visible spectral range to ensure lower-cost equipment to record (Fuller et al. 2003). Upon excitation, the emissive triplet state is quenched (deactivated) by the triplet molecular oxygen ground state (Bacon and Demas 1987). Therefore, sensor layers in oxygen-free water show the strongest fluorescence intensity I_0 . The quenching effect is a function of temperature and oxygen concentration (Baleizão et al. 2008). At constant temperature, the Stern–Volmer plots show a nonlinear behavior for most oxygen indicator dyes (Carraway et al. 1991b). This behavior can be described by a modified Stern–Volmer equation based on a two-site quenching model (Carraway et al. 1991b, Demas et al. 1995):

$$\frac{I_0}{I} = \left[\frac{f}{1 + K_{SV}[O_2]} + (1 - f) \right]^{-1} \quad (1)$$

Equation 1 is a simplified form of the general two-site model assuming a constant non-quenchable second fraction, where I is the fluorescence intensity, K_{SV} is the Stern–Volmer constant, $[O_2]$ is the oxygen concentration, and f and $(1-f)$ reflects the quenchable and non-quenchable fraction of the fluorescent dye, respectively (Carraway et al. 1991a).

Preparation of the oxygen sensing matrix Most of such optical sensor materials consist of a dye incorporated in a polymeric matrix that serves as a solvent for the dye, as a solid support, and as a selectively permeable membrane. The sensor foil for oxygen was prepared in two steps: 20 mg of platinum (II) 5,10,15,20-tetrakis(2,3,4,5,6-pentafluorophenyl)porphyrin purchased from Porphyrin Systems (www.porphyrin-systems.de) and 1.98 g of polystyrene obtained from Sigma-Aldrich (www.sigmaaldrich.de) were dissolved in 18 g of toluene purchased from Sigma-Aldrich. The “cocktail” was stirred overnight and knife-coated onto a 100 μm polyester support (Melinex 505, www.puetz-folien.com). After solvent evaporation at ambient air, the resulting sensor foils are in the range of 10 μm thickness. Thickness is an important parameter especially for oxygen sensing because lateral diffusion of the analyte in the sensitive film may cause limitations (Liebsch et al. 2000). Schroeder et al. (2007) showed that this complex has an increased photostability due to the electron-withdrawing pentafluorophenyl substituents of the porphyrin ligands. Moreover, in marine experiments under water-saturated conditions, no significant change in the calibration function was monitored within a period of 18 months (Precht et al. 2004), which implies that the dye neither was bleached nor diffused out of the foil in substantial amounts. For the latter reason, possible phytotoxicity of the dye was not a

prime issue because the foil itself was stable and this was not assessed here.

Calibration of the oxygen sensing matrix The calibration of the sensor was performed at room temperature (21°C) with determination of the fluorescence intensity of the dye at several oxygen concentrations. We conducted the calibration under conditions similar to the experiments. The container was filled with the same porous medium and flushed with water of different oxygen concentrations. The fluorescence intensity of the oxygen indicator versus oxygen concentration is displayed in Fig. 1. The “two-site model” described in Eq. 1 was used to fit the data. The model fits the experimental data very well. Moreover, f and K_{SV} are determined from the calibration points by rearranging Eq. 1 (Glud et al. 1996). The calculated values for f and K_{SV} were 0.37 and 0.19 mg l⁻¹, respectively.

2.2 Neutron radiography

Neutrons and protons are constituents of the atomic nuclei. Whereas protons have a positive net charge, neutrons are electrically neutral. Because of their zero net charge neutrons do not interact with electrons, but with atomic nuclei, and have highly different attenuation coefficients even for elements of similar atomic number or isotopes of one element. This is in contrast to X-rays, which interact with electronic charge and therefore the interaction increases with atomic number. Whereas X-ray imaging strongly responds to heavy elements, NR in comparison is sensitive mainly to hydrogen and hydrogen-bearing materials. This dominant sensitivity to ¹H in soil systems results in a good visualization of water and roots, which mainly consist

of water (Moradi et al. 2009). The high spatial and temporal resolution of neutron radiography permits the visualization of changes at the soil–root interface without disturbance of the gradients. Information about mass and thickness of neutron attenuation components can be derived by comparing the transmitted and unperturbed neutron beam. Neutron radiography techniques are based on the exponential law (similar to the Lambert–Beer law) of attenuation of radiation passing through matter, in our case composed of grains, water, and air (e.g., Oswald et al. 2008):

$$I(x, z) = I_0(x, z) \times e^{-\Sigma L(x, z)} \quad (2)$$

where (x, z) is the plane perpendicular to the neutron beam direction y and $I(x, z)$ is the detected neutron flux after an incident neutron flux $I_0(x, z)$ passes through material of thickness $L(x, z)$ with a neutron attenuation coefficient of Σ .

Image analysis Neutron radiographs show the neutron penetration through the sample in the direction of the beam. Neutrons can be absorbed or scattered, but they can also pass the sample undisturbed. With increasing water content, in soil there is a much higher probability for neutron scattering than for neutron absorption (Hassanein 2006). Scattered neutrons may reach the detector and then increase the image intensity recorded, which leads to an underestimation of the neutron attenuation or ultimately water mass density (Moradi et al. 2009). In our study, the transmission images were first processed by the quantitative neutron imaging (QNI) algorithm (Hassanein 2006). It corrects the images for scattering, beam fluctuations, detector non-uniformities, and camera noise using the open beam image without the sample and the image without beam (dark current). After correction, the intensity of the transmitted neutron beam through the sample is described by the exponential law taking into account the individual attenuation coefficients of the sample:

$$-\log \left[\frac{I(x, z)}{I_0(x, z)} \right] = \Sigma_{\text{cont}} L_{\text{cont}}(x, z) + \Sigma_{\text{H}_2\text{O}} L_{\text{H}_2\text{O}}(x, z) + \Sigma_{\text{soil}} L_{\text{soil}}(x, z) \quad (3)$$

The subscript “cont” refers to the container, H₂O to water, and “soil” to the dry soil (quartz). The attenuation coefficient of water is calculated by QNI software. Then, rearranging Eq. 3, we calculate $L_{\text{H}_2\text{O}}(x, z)$. The water content θ is defined as:

$$\theta(x, z) = \frac{L_{\text{H}_2\text{O}}(x, z)}{L_{\text{tot}}(x, z)} \quad (4)$$

where $L_{\text{tot}}(x, z)$ is the sample thickness. Equation 4 yields the water content averaged along the beam direction

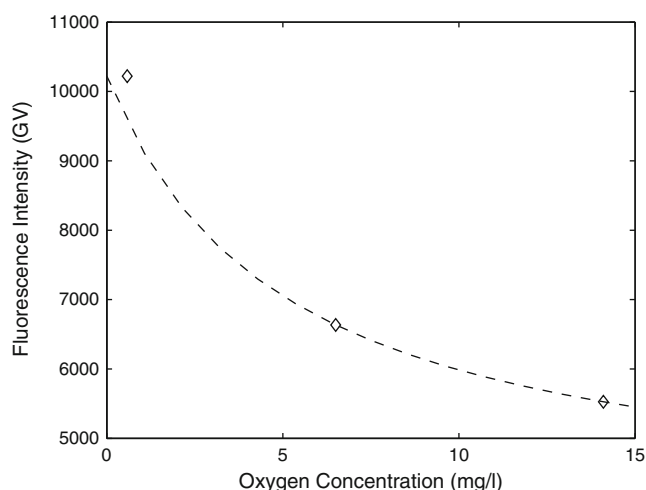


Fig. 1 Fluorescence intensity as a function of dissolved O₂ concentration in water. The dotted line represents the fit with the modified Stern–Volmer equation (Eq. 1)

(Carminati and Flühler 2009). For this study, we expect that the two-dimensional average of the water content is sufficient to compare the data with the fluorescence images.

2.3 Experimental setup

The first part of the experiment involved imaging water content and oxygen concentrations in a homogeneously packed glass bead medium, without plant roots, as a relatively simple system. By that, we could test the two imaging methods and highlight the influence of the water content distribution on oxygen diffusion at room temperature (21°C). The containers had internal dimensions of $0.1 \times 0.2 \times 0.05$ m and were filled with glass beads (from Carl Roth, Germany; www.carl-roth.de). The glass beads had a grain size distribution of 0.75–1.0 mm. The containers were open to atmosphere at the top and had a port at the bottom for drainage and imbibition. The container chosen for containing a fully saturated porous medium was filled up with water; afterward, dry glass beads were slowly moved in from a hopper. The porosity was about 33%. The container to represent unsaturated conditions was filled up with dry glass beads and then was wetted by raising the water table via the port at the bottom. The porosity was about 30%, which was determined by the known glass density, the inner volume of the container, and the dry mass of glass beads. The water content was estimated gravimetrically and was lower than for the saturated setup. The same porous medium was used to conduct a simple drainage experiment. The water content distribution in a container filled with the same glass beads was calculated. The average water content in the drained upper part of the container was 0.02 and 0.28 in the saturated lower part. In general, the grain size distribution affects the water content distribution, but also the shape and size of the capillary fringe.

The next step targeted plant experiments in thin containers. The size was a compromise between achieving a good contrast in the image (the thinner the better) and providing adequate space for germination and root development. The sensor foils for O_2 were installed inside on the container walls. These types of sensors show short response times to changes in the environment in order of minutes or less and seem completely reversible. The containers were made of glass because the optical transparency in the visible range was required for fluorescence imaging. Additionally, the glass chosen has a low attenuation for neutrons resulting in containers that were also transparent for the incoming neutrons. Firstly, we performed a series of experiments with plants in four containers that had dimensions of $0.1 \times 0.1 \times 0.01$ m and were open to atmosphere at the top. The containers were filled homogeneously with quartz composed of 85% sand, 10% silt, and 5% clay. White lupine (*Lupinus*

albus L.) or blue lupine (*Lupinus angustifolius* L.) seeds were planted in the container.

The last set of experiments then looked at the most complex situation, with a plant grown in a structured soil in a similar way. Into a homogeneous quartz sand mixture (grain size distribution of 200–2,000 μm), we built a layer of finer-grained quartz sand (grain size distribution of 1–250 μm). The layer of finer sand was designed to block the lower compartment during drying from quick links to the gas phase in the upper part and thus the atmosphere. The two containers with internal dimensions of $0.15 \times 0.15 \times 0.01$ m were planted with white lupine (*L. albus* L.). The experiments were performed at the neutron radiography facility CONRAD at the Helmholtz Centre Berlin (HZB), Germany. The facility is located at the reactor-based neutron source BER 2 and consists of a special beam collimator, a two-dimensional neutron detector, and a cooled CCD camera (Fig. 2, left, A). A mobile plant growth cabinet (Vötsch, Balingen, Germany) was placed at the facility for the experiments. All plants were grown in containers starting from a single seed placed into the sand surface and watered to start germination. The plants were grown for 2 weeks with a daily photoperiod of 14 h (starting at 6 am to 8 pm), day temperature of 25°C, night temperature of 19°C, and relative humidity of about 60%. After a growing period of 2 weeks, the samples were wetted from bottom. During five day and night cycles, we took radiographies and imaged the root growth, water content, and oxygen distribution. For each sample, an image, or image set, was taken every 3 h during the day–night cycle to monitor intraday variations. To have an adequate illumination of the sample, the container usually is chosen smaller than the diameter of the beam. We could enable imaging the 0.15×0.15 -m containers, though larger than the maximum field of view of just 0.1×0.1 m at CONRAD, by scanning nine single images of an overlapping 3×3 array. In image processing, the images were stitched together by applying the ImageJ plugin “TrakEM2” resulting in an image covering the full size of the container.

At the neutron imaging station CONRAD, a special setup was built inside the shielded area to enable automatic positioning of the equipment during operation (see Fig. 2, left). The samples were placed onto a positioning system to be able to move quickly and precisely between the position of the neutron radiography setup and the position for optical fluorescence imaging. At position A (see Fig. 2), the beam was transmitted through the sample, transferred into light impulses at the neutron detector, and a built in CCD camera recorded the resulting image. In essence, gray value images are taken with values proportional to the neutron flux behind the sample. The intensity of the neutron flux was recorded as a two-dimensional transmission map (image) for the sample positioned orthogonal to the beam

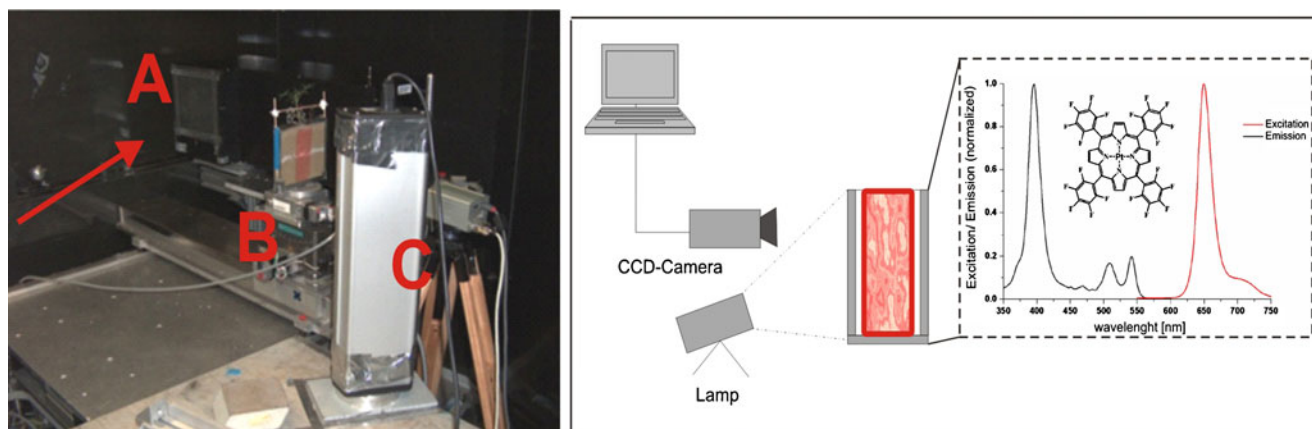


Fig. 2 Coupled setup for neutron radiography and fluorescence foil imaging (*left*). *A* Position of sample in front of neutron detector with neutron beam direction as indicated by the arrow; *B* sample on

positioning table, rotated for optical imaging from the right; *C* UV lamp and CCD camera. The *sketch on the right* outlines in detail the fluorescence foil imaging setup

(Strobl et al. 2009), and therefore, the resulting image stores a water content value that is integrated in the beam direction over the thickness d of the container (Moradi et al. 2009). Though also a three-dimensional tomography of these containers is possible (Esser et al. 2010), this two-dimensional imaging setup gave a higher temporal resolution that was essential for capturing the dynamics of the oxygen distribution changes. The sketch (see Fig. 2 right) describes in more detail the components of the fluorescence imaging part. The containers were illuminated by a UV lamp (typ 215L, Fa. Peqlab). The fluorescence was recorded by a CCD camera (type DX 4C-285, Kappa Optoelectronics) that provided a resolution of $1,392 \times 1,040$ pixels with 12-bit digitization. The CCD chip was cooled to -15°C to avoid thermal noise. The distance between the cameras and the container was 0.3 m. The UV lamp was placed next to the CCD camera at the height of the sample to ensure homogeneous illumination and to prevent shadows. The dye was excited with high efficiency at the Soret band (~ 400 nm), and the emission maximum was located at ~ 650 nm. The excitation light and emission of the luminescent compound were separated by a 500-nm long-pass filter (www.stemmer-imaging.de). For neutron radiography, we used a field of view of 100×110 mm, obtaining an image of $2,048 \times 2,048$ pixels with pixel size of 0.054 mm. Fluorescence foil imaging achieved a camera resolution of 0.21 mm with a field of view of 258×221 mm.

3 Results

The experiments investigated water content and oxygen concentrations in artificial porous media. We report in the following the results (a) in a homogeneously packed glass bead medium without plant roots, (b) in a homogeneously

packed quartz sand with a plant, and (c) for the most complex situation, with a plant grown in a structured soil, with an intermediate layer of finer-grained quartz sand.

3.1 Homogeneous porous media

We mapped the oxygen concentration in a container homogeneously packed with glass beads of uniform size. Images were taken for saturated conditions and for unsaturated conditions, but close to saturation (Fig. 3; after 2 and 4 h of diffusion). We monitored the oxygen content and observed reduced oxygen transport for unsaturated conditions. By comparing fully saturated conditions with the situation close to saturation, but containing trapped gas bubbles, we observed that small amounts of trapped gas can retard the movement of dissolved oxygen. A simple relationship to estimate the transport distance of diffusion is given by:

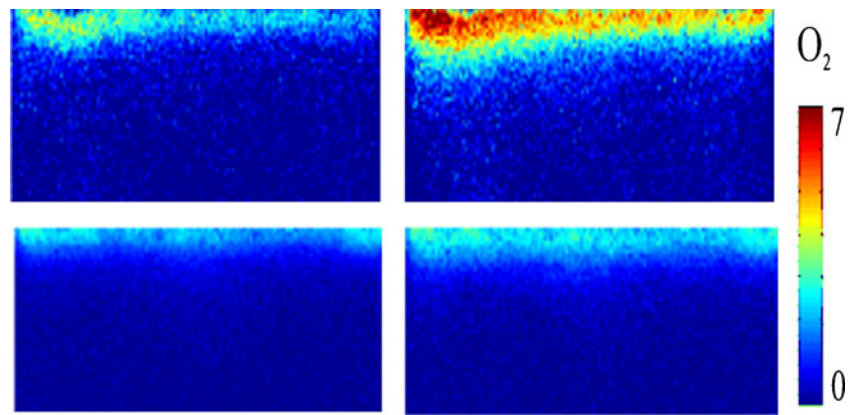
$$x^2 = 2Dt$$

where $D = 1.2 \times 10^{-9} \text{ m}^2 \text{ s}^{-1}$ is the diffusion coefficient of oxygen in water accounting for the tortuosity of a porous medium (Huang et al. 2003). The resulting distance 0.588 cm fitted perfectly the measured one given by the image (0.576 cm).

3.2 Homogeneous porous media with plant roots

The distribution of oxygen around the roots of a 14-day-old lupine plant was measured, quasi-simultaneously with the soil water content distribution obtained by neutron radiography (Fig. 4). The oxygen-sensitive foil covered the central part of the container only. The left image in Fig. 4 shows the corrected neutron radiogram of the container, which captured the root structure, i.e., after 14 days of root

Fig. 3 Oxygen concentration (milligrams per liter) after 2 h (left) and 4 h (right) of oxygen diffusion from top in contact with atmosphere into the anaerobic water: in partly unsaturated glass beads, initially filled with nitrogen gas (bottom images), and in fully saturated glass beads (top images)



development. The oxygen-sensitive foil covers main parts of the root system including tap root and lateral roots (as is illustrated in Fig. 4). These roots are clearly visible in the neutron radiogram. The observed oxygen patterns seem patchy and inhomogeneous. The combination with neutron radiography allows to judge if this was caused by plant roots growing inside the porous medium.

By superimposing the segmented roots from neutron radiography on top of the fluorescence image, we could spatially match roots and oxygen concentration and highlight that the oxygen deficits occur around the roots. The oxygen maps indicated that the depletion of oxygen extended to distances of 2.94 mm from the root surface (see Fig. 4, right). This could be clearly identified by the spatial resolution of 0.21 mm.

The combined setup enables us to capture day and night cycles concerning water content and oxygen distribution (Fig. 5). Uptake of water varied significantly between day and night. During a night cycle, we monitored almost no change in the average water content, whereas during the following day, the average water content dropped from 0.35

to 0.25. The related patterns of oxygen depletion were also monitored during a day and during a night (see Fig. 5, upper and lower row). Both represent conditions close to water saturation because the samples were wetted directly before starting to monitor the oxygen depletion. The depletion zones started at the location of the tap root and largest lateral roots and extended to the whole root zone after a few hours. Oxygen depletion was similar at day and night, though it seemed a bit smaller and more diffuse during night time.

The sensor foil consisting of polymeric substances influenced the neutron radiography images. The additional neutron attenuation of the foil created a higher background, which resulted in seemingly higher water content. Thus, at the area of the sensor foil, we observed initially a slightly higher water content than the calculated average water content (about $\theta=0.05$). We have corrected for that effect by using a comparison of a reference zone inside and outside the area of the foil.

Figure 6 shows a time series of oxygen deficit dynamics over 16 h after watering the sample. We observed that the extension of oxygen depletion by respiration was not limited to the rhizosphere. The depletion zones developed and increased from local, patchy depletion patterns, close to the roots, to a diffuse pattern over a larger area after several hours. From day 14 at 3 pm to 10 pm, the upper part of the container shows a slight re-aeration, limited to the top centimeters of sand. This could be related to decreasing water saturation by water losses due to evapotranspiration. Oxygen depletion was still increasing due to ongoing respiration activities in the lower part of the container for some time. Then at day 15 at 7:20 am (see Fig. 6), comparing with the former image at day 14 at 9:50 pm, we observed that the oxygen concentration increased in all parts of the container, and the oxygen depletion zone was much smaller and weaker. Both images are separated by a night cycle, meaning little transpiration, but ongoing respiration activity. By additionally taking into account the neutron radiography images, there is no change in the

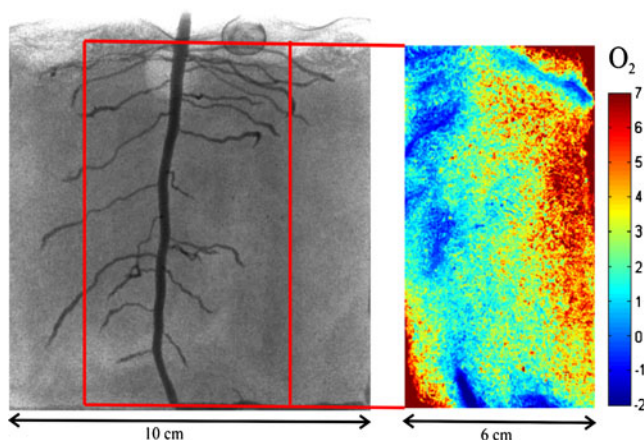


Fig. 4 Container with lupine in a quartz sand captured by NR (corrected raw image, left) and position of the oxygen-sensitive foil. Oxygen concentration in the rectangular area 2.5 h after wetting (right)

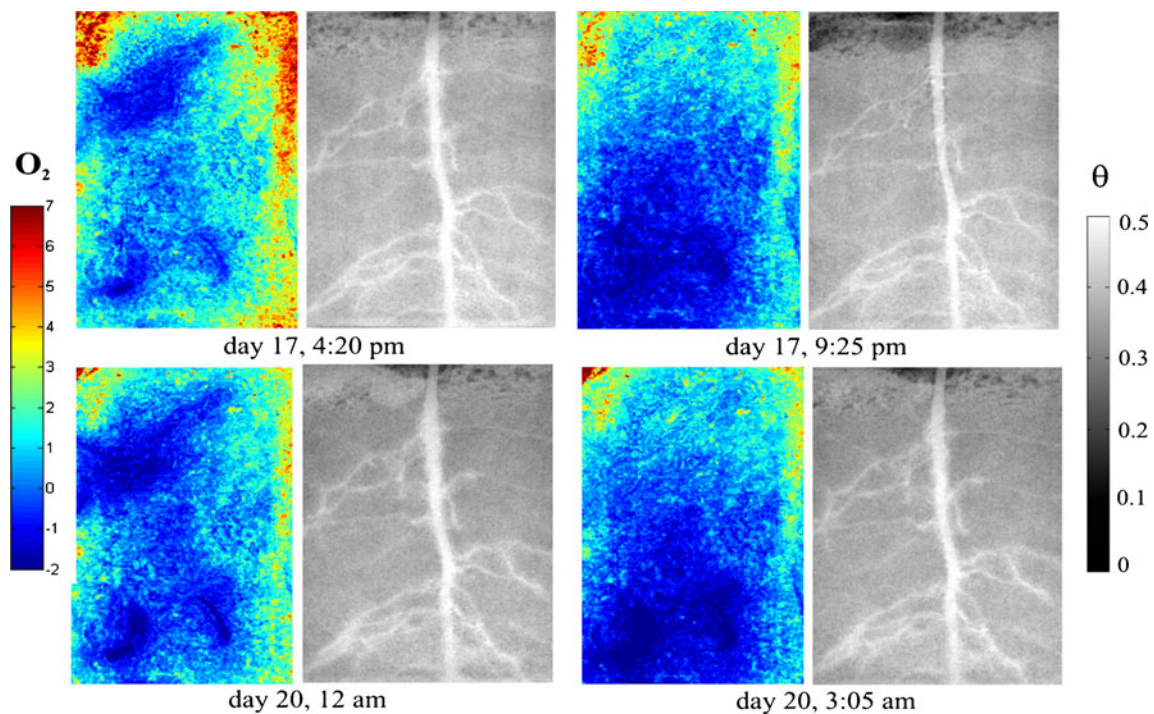


Fig. 5 Time series of water content distribution in homogeneous porous media with lupine plant (*left*) and two according time series of oxygen patterns for the same sample (*right*) in milligrams per liter; *upper row* shows a day cycle and *lower row* a night cycle

overall water content distribution, which is still high with an average water content of 0.37 for both images.

3.3 Layered porous media with plant roots

The heterogeneous structure of soil is clearly visible in the water content images (Fig. 7). We observed a clear separation of water contents during drying of the soil that strongly reduced the water content in the upper part. This is

essentially caused by the built-in clay layer in the middle of the container. The finer-grained clay holds more water than the surrounding soil, which resulted in higher water contents. Moreover, this high water content in the clay layer limits the downward oxygen diffusion, which results in a generally low oxygen concentration compared to the coarser-grained sand above, exposed to oxygen in the atmosphere. Since any oxygen exchange between the lower part and the atmosphere was limited by the clay layer, it is

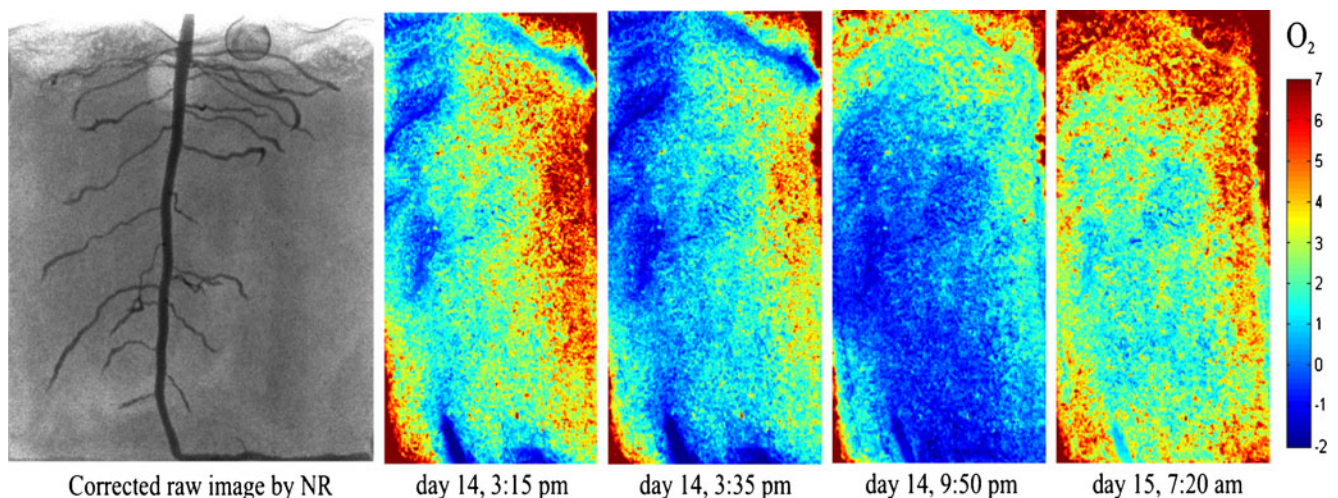


Fig. 6 Time series of oxygen patterns (concentration in milligrams per liter) induced by plant roots in a container with homogeneous porous medium

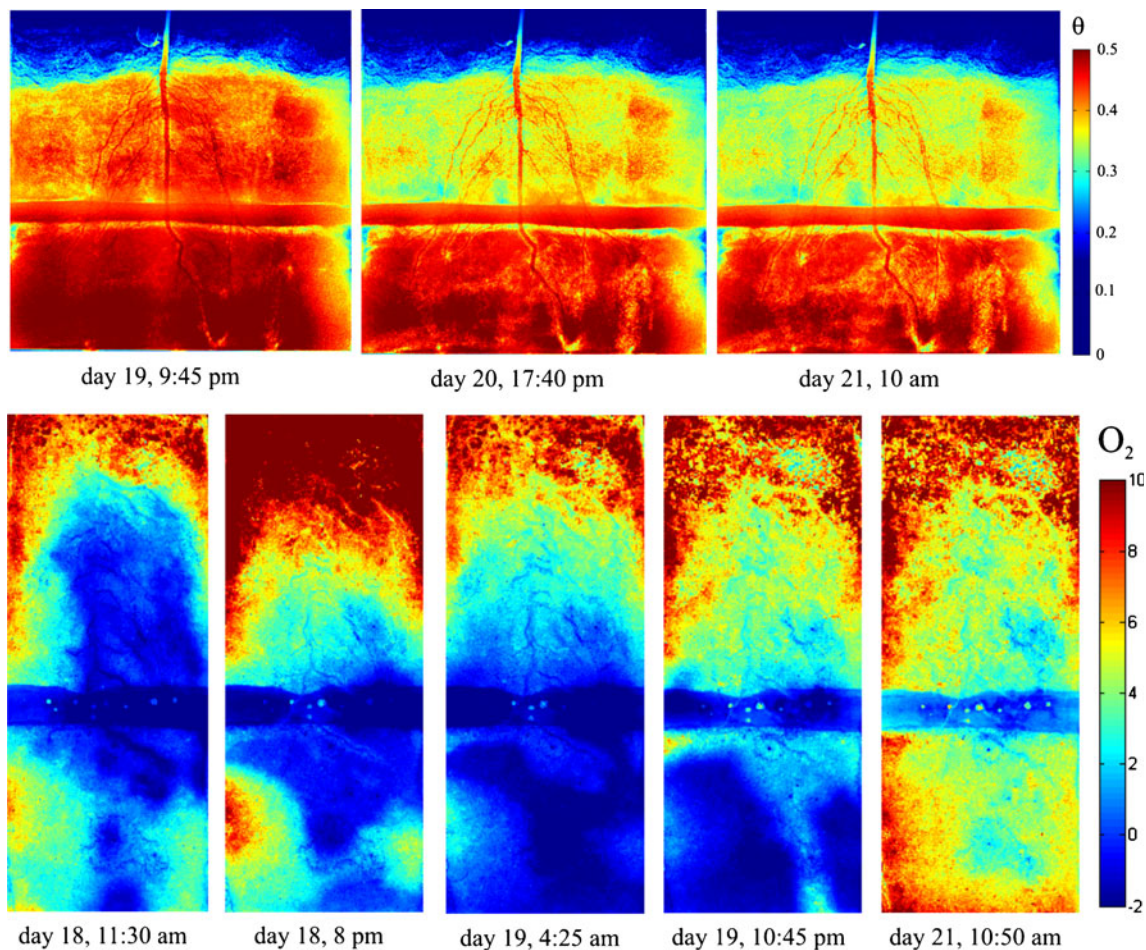


Fig. 7 Time series of water content distribution in a container with a layer of fine quartz sand (*upper row*); corresponding distribution of oxygen (in milligrams per liter) in the central area of the container where the oxygen sensing foil is installed (*lower row*)

reasonable to assume that any change in oxygen concentration in the lower part of the container was linked to the activity of roots and soil microorganisms, plus potentially lateral diffusion of oxygen from the roots. The position of the oxygen-sensitive foil is comparable with Fig. 4. The foil here also covered most of the main part of the root system including the tap root and the laterals.

We observed a decrease in water content in both compartments from days 19 to 21 (see Fig. 7, upper row). The evapotranspiration was stronger in the upper compartment compared to the lower one also because the root mass was much higher in the upper part. It can be concluded that water loss in the lower part was directly linked to root water uptake. The water content distributions on days 20 and 21, respectively, showed reduced water contents around the roots compared to bulk soil. Comparing water content distributions from days 19 to 20, again we could observe strong day and night variations. Due to physiological reasons (compare Section 3.2), we expected and monitored reduced root water uptake rate during night. The average water loss in the samples with white lupine was 0.46 g/h during the

photoperiod and 0.21 g/h during night. Containers planted with blue lupines had an average water loss of 0.42 and 0.23 g/h during night. The averaged water content in the lower compartment dropped from 0.50 to 0.46 between day 19 and day 21, respectively.

After rewetting (early morning of day 18), oxygen deficit zones formed, induced by respiration activity of the plant roots (see Fig. 7, lower row). The time series showed a decreased oxygen concentration at the positions of the roots. Moreover, the lower compartment showed increasing oxygen deficits. The oxygen concentrations decreased until morning of day 19, though in the upper part oxygen was re-entering already at day 18. Then, in the morning of day 19 at 10:45 pm, oxygen re-entered also in the lower part, along the position of the tap root. However, as it looks like this occurred starting from the bottom of the container, as visible between 4:25 am and 10:45 am. This correlated with a hole in the bottom at the right-hand side for wetting the sample. Thus, the seal seemed to have failed probably because the tap root was growing into this opening. Without this connection at the bottom, the oxygen

deficit in the lower part may have lasted longer. The clay layer itself was depleted in oxygen until the last image at day 21.

The upper compartment was open to atmosphere, and oxygen diffusion was only limited in case of high water saturation. Neutron radiography provided the information that the root density in the upper compartment was higher than in the lower one. Due to higher root density, the water uptake was stronger than in the lower part. The oxygen deficit in the upper compartment increased until day 18 at 11.30 am due to high water content. In the following day cycle (starting with day 18 at 8 pm), we observed an oxygenation in the upper compartment, which could be explained by the decrease of the soil water content and resulting recovery of connected air phase and exchange of oxygen between the soil pores and the atmosphere. The slight decrease in oxygen in the upper compartment on day 19, 4:25 am could be due to the redistribution of soil water following the pausing of root water uptake. The next day cycle restarts the transpiration and results again in re-aeration of the upper part of the container.

4 Discussion

Within the first set of experiments, we showed that knowledge of water content distribution is important concerning the transport of oxygen. The simple setup with partly saturated porous media visualized the strong coupling of both parameters. The mass transfer of oxygen for fully saturated reaching farther into the porous medium than in the presence of trapped gas bubbles (see Fig. 3). Oxygen will transfer from mobile aqueous phase and gas phase by diffusion. Oxygen concentration gradients between both phases are the driving force until equilibrium is reached (Balcke et al. 2007). At the gas–liquid interface, the mass transfer is controlled by the rate of diffusion of the oxygen across a thin layer assumed to exist at the phase interface (Donaldson et al. 1997). Though oxygen diffusion in the gas phase is much higher than in water, the trapped gas bubbles seem to act rather as a storage retarding the oxygen front than as a preferential pathway. This is different for connecting gas phase as can be seen in the other experiments.

The second set of experiments proved that the formation of oxygen deficits can be directly related to root activity, more specific to root respiration. The combination of neutron radiography and fluorescence dye imaging is able to simultaneously capture the concentrations of oxygen and water content. Moreover, the characteristics of the oxygen sensor give us the possibility to monitor dynamics of these parameters in situ. This enables us to capture day and night variations in the root zone. Our experiments indicate that oxygen distribution in soils is highly variable in time and

space and is related to water content and roots (see Fig. 5 and 6). Whereas only green tissues photosynthesize, all tissues respire, and they do so 24 h a day. Thus, this oxygen depletion is to be expected and could now be monitored in more detail by the presented imaging approach.

The average water content from evening of day 14 to the morning of day 15 decreased slightly (see Fig. 6). It could be possible that the air phase just can form a connected gas-filled pore network, probably by the larger pores present in the soil. This may not show up in the water content images, especially since the averaging of the two-dimensional neutron radiography over the thickness of the container will make this difficult to detect. If so, there is a connected network for the quick gas phase diffusion of oxygen, and oxygen from the atmosphere will replenish the oxygen in the soil. Thus, this could explain the increased oxygen values in spite of ongoing oxygen consumption due to respiration activity. Possibly even this threshold was already reached in the evening, but some more time was needed for the oxygen transfer from the atmosphere to greater depth to result in the observed changes. To sum up, oxygen depletion zones will result, provided that gas phase re-supply of oxygen is limited due to high water content. Further investigations are needed to differentiate the effect of plant roots and soil structure of oxygen movement in soil.

The third, most complex setup presented in this study confirms the applicability for the combined approach. The experimental setup was able to visualize patches of different grain size distributions and roots. Natural soils are highly heterogeneous, and the third set of experiments showed the application also to a structured soil (see Fig. 7). Again, these experiments showed the strong interrelation between root water uptake forced by transpiration activity, oxygen depletion induced by root respiration, and soil structure. Besides mass transfer of oxygen between liquid and gas phase had an influence on the observed patterns. Closer experiments on the interaction of oxygen change and water content distributions are necessary to better understand how and where plants react on oxygen limitations.

Combining fluorescence dye imaging with neutron radiography, which is highly sensitive for water, shows a minor drawback. The supporting foils for fluorescence imaging are visible in neutron radiography because the foils consist of an organic polymer that absorbs neutrons efficiently, similar to water. A background correction for the additional attenuation by the foil was applied, and may be necessary for some setups and some facilities to give accurate water content values and may not be necessary for others. Also, the fluorescence dye layer could be manufactured on different substrates, e.g., directly on glass, which should result in a much lower background.

Combining neutron radiography and fluorescence imaging could be most effective by targeting a spatial resolution

around 200 μm , i.e., less than the possible 54- μm resolution achieved as described in this study. This could allow for imaging larger areas, and the obtained water content distributions in combination with fluorescence dye imaging should still be sufficient to allow a similar interpretation as the one presented here.

The experiments have demonstrated that by this novel methodology, it is possible to monitor, with high spatial and temporal resolution, the dynamics of oxygen and water in the rhizosphere as an interface between soil and roots. The rhizosphere is accepted to be an area around the roots where most of biological activities occur in soil (Hinsinger et al. 2009). Moreover, the rhizosphere is home to most of the soil microorganisms that feed on decomposing root residue and root exudates on organic materials. Besides root respiration, microbial induced oxygen consumption may have had a major effect on the observed oxygen distributions.

Both imaging methods work in quantitative way and gave meaningful results. In spite of the averaging character of two-dimensional radiography and the sensor foil being in contact with the lateral soil surface, the results demonstrate that the processes and dynamics inside soil can be visualized. Thus, zones of oxygen depletion could be mapped and linked to root respiration, plant transpiration, and water content distribution.

5 Conclusions

Within this study, we presented a methodology with fluorescence dye imaging to map the changes in oxygen concentration in soil–root systems, which is also applicable to natural soils with plants. As a second step, we already coupled the optical imaging with highly sensitive neutron radiography to monitor the water content distribution of the same sample at nearly the same time. The formation of the observed oxygen depletion zones is highly sensitive to the water content distribution because zones with high water saturation are able to block the supply of oxygen via the soil gas phase. Therefore, the major advantage of this combined approach is that the oxygen dynamics and changing patterns of oxygen depletion and oxygen availability can be interpreted in a more profound and meaningful way.

Generally, the presented fluorescence foil imaging method is applicable to porous media with focus on soil structure, sediments, or microbes. Moreover, one great advantage of fluorescence dye imaging is the broad range of adaptability. Besides oxygen, temperature, pH, or CO_2 gradients are other important biogeochemical parameters. Concerning vital parameters of plants, e.g., pH is also of interest, because of the strong correlation between pH, water, oxygen, and root activity in the rhizosphere (a) water

uptake by plant roots is negatively influenced by increasing pH, (b) anaerobic conditions cause a decrease in respiration activity and this in turn increases the pH (Tournaire-Roux et al. 2003), and (c) root exudates directly modify pH and thus trace metal availability. The fluorescence dye imaging method has the potential for visualization of pH and oxygen, and possibly other parameters, in the root zone, soils, or sediments, even at the same time, which could be achieved by fluorescent multisensing (Nagl and Wolfbeis 2007). As presented here, also water content can be simultaneously mapped, if optical imaging is combined with neutron radiography.

Acknowledgments We acknowledge funding from the DFG priority program 1315 “Biogeochemical interfaces in soil”, under contract OS 351/1-2, and the support from the Helmholtz Centre for Environmental Research—UFZ, departments Hydrogeology and Soil Physics.

References

- Amao Y, Okura I (2000) An oxygen sensing system based on the phosphorescence quenching of metalloporphyrin thin film on alumina plates. *Analyst* 125:1601–1604
- Bacon JR, Demas JN (1987) Determination of oxygen concentrations by luminescence quenching of a polymer-immobilized transition-metal complex. *Anal Chem* 59:2780–2785
- Balcke GU, Meenken S, Hoefer C, Oswald SE (2007) Kinetic gas–water transfer and gas accumulation in porous media during pulsed oxygen sparging. *Environ Sci Technol* 41:4428–4434
- Baleizão C, Nagl S, Schaferling M, Berberan-Santos MN, Wolfbeis OS (2008) Dual fluorescence sensor for trace oxygen and temperature with unmatched range and sensitivity. *Anal Chem* 80:6449–6457
- Blossfeld S, Gansert D (2007) A novel non-invasive optical method for quantitative visualization of pH dynamics in the rhizosphere of plants. *Plant Cell and Environment* 30:176–186
- Blossfeld S, Gansert D, Thiele B, Kuhn AJ, Lösch R (2011) The dynamics of oxygen concentration, pH value, and organic acids in the rhizosphere of *Juncus* spp. *Soil Biology Biochemistry* 43:1186–1197
- Borisov SM, Vasylevska AS, Krause C, Wolfbeis OS (2006) Composite luminescent material for dual sensing of oxygen and temperature. *Advanced Functional Materials* 16:1536–1542
- Brune A, Frenzel P, Cypionka H (2000) Life at the oxic–anoxic interface: microbial activities and adaptations. *Fems Microbiol Rev* 24:691–710
- Carminati A, Flühler H (2009) Water infiltration and redistribution in soil aggregate packings. *Vadose Zone J* 8:150–157
- Carraway ER, Demas JN, Degraff BA (1991a) Luminescence quenching mechanism for microheterogeneous systems. *Anal Chem* 63:332–336
- Carraway ER, Demas JN, DeGraff BA, Bundt M (1991b) Photo-physics and photochemistry of oxygen sensors based on luminescent transition-metal-complexes. *Anal Chem* 63:337–342
- Deinert MR, Parlange JY, Steenhuis T, Throop J, Unlu K, Cady KB (2004) Measurement of fluid contents and wetting front profiles by real-time neutron radiography. *J Hydrol* 290:192–201
- Demas JN, Degraff BA, Xu WY (1995) Modeling of luminescence quenching-based sensors—comparison of multisite and nonlinear gas solubility models. *Anal Chem* 67:1377–1380

- Donaldson JH, Istok JD, Humphrey MD, O'Reilly KT, Hawelka CA, Mohr DH (1997) Development and testing of a kinetic model for oxygen transport in porous media in the presence of trapped gas. *Ground Water* 35:270–279
- Esser HG, Carminati A, Vontobel P, Lehmann EH, Oswald SE (2010) Neutron radiography and tomography of water distribution in the root zone. *J Plant Nutr Soil Sci* 173(5):757–764
- Fuller ZJ, Bare WD, Kneas KA, Xu WY, Demas JN, DeGraff BA (2003) Photostability of luminescent ruthenium(II) complexes in polymers and in solution. *Anal Chem* 75:2670–2677
- Glud RN, Ramsing NB, Gundersen JK, Klimant I (1996) Planar optodes: A new tool for fine scale measurements of two-dimensional O₂ distribution in benthic communities. *Mar Ecol-Prog Ser* 140:217–226
- Glud RN, Santegeerts CM, De Beer D, Kohls O, Ramsing NB (1998) Oxygen dynamics at the base of a biofilm studied with planar optodes. *Aquatic Microbial Ecology* 14:223–233
- Hassanein RK (2006) Correction methods for the quantitative evaluation of thermal neutron tomography. Dissertation, ETH Zurich
- Hinsinger P, Bengough AG, Vetterlein D, Young IM (2009) Rhizosphere: biophysics, biogeochemistry and ecological relevance. *Plant Soil* 321:117–152
- Huang WE, Oswald SE, Lerner DN, Smith CC, Zheng C (2003) Dissolved oxygen imaging in a porous medium to investigate biodegradation in a plume with limited electron acceptor supply. *Environ Sci Technol* 37:1905–1911
- Jensen SI, Kuhl M, Glud RN, Jorgensen LB, Prieme A (2005) Oxidic microzones and radial oxygen loss from roots of *Zostera marina*. *Mar Ecol-Prog Ser* 293:49–58
- Kardjilov N, Hilger A, Manke I, Strobl M, Treimer W, Banhart J (2005) Industrial applications at the new cold neutron radiography and tomography facility of the HMI. *Nucl Instrum Meth A* 542:16–21
- Klimant I, Meyer V, Kuhl M (1995) Fiberoptic oxygen microsensors, a new tool in aquatic biology. *Limnol Oceanogr* 40:1159–1165
- Lehmann E, Pleinert H, Williams T, Pralong C (1999) Application of new radiation detection techniques at the Paul Scherrer Institut, especially at the spallation neutron source. *Nucl Instrum Meth A* 424:158–164
- Liebsch G, Klimant I, Wolfbeis OS (1999) Luminescence lifetime temperature sensing based on sol–gels and poly(acrylonitrile)s dyed with ruthenium metal-ligand complexes. *Advanced Materials* 11:1296–1299
- Liebsch G, Klimant I, Frank B, Holst G, Wolfbeis OS (2000) Luminescence lifetime imaging of oxygen, pH, and carbon dioxide distribution using optical sensors. *Applied Spectroscopy* 54:548–559
- Luster J, Gottlein A, Nowack B, Sarret G (2009) Sampling, defining, characterising and modeling the rhizosphere—the soil science tool box. *Plant Soil* 321:457–482
- Matsushima U, Herppich WB, Kardjilov N, Graf W, Hilger A, Manke I (2009) Estimation of water flow velocity in small plants using cold neutron imaging with D₂O tracer. *Nucl Instrum Meth A* 605:146–149
- Mills A, Lepre A (1997) Controlling the response characteristics of luminescent porphyrin plastic film sensors for oxygen. *Anal Chem* 69:4653–4659
- Moradi AB, Conesa HM, Robinson B, Lehmann E, Kuehne G, Kaestner A, Oswald S, Schulin R (2009) Neutron radiography as a tool for revealing root development in soil: capabilities and limitations. *Plant Soil* 318:243–255
- Moradi AB, Carminati A, Vetterlein D, Vontobel P, Lehmann E, Weller U, Vogel HJ, Oswald SE (2011) Three-dimensional visualization and quantification of water content in the rhizosphere. *New Phytol*. doi:10.1111/j.1469-8137.2011.03826.x
- Nagl S, Wolfbeis OS (2007) Optical multiple chemical sensing: status and current challenges. *Analyst* 132:507–511
- Nagl S, Baleizao C, Borisov SM, Schaferling M, Berberan-Santos MN, Wolfbeis OS (2007) Optical sensing and imaging of trace oxygen with record response. *Angew Chem Int Edit* 46:2317–2319
- Oswald SE, Menon M, Carminati A, Vontobel P, Lehmann E, Schulin R (2008) Quantitative imaging of infiltration, root growth, and root water uptake via neutron radiography. *Vadose Zone J* 7:1035–1047
- Precht E, Franke U, Polerecky L, Huettel M (2004) Oxygen dynamics in permeable sediments with wave-driven pore water exchange. *Limnol Oceanogr* 49:693–705
- Schreiber C (2010) Rhizosphere dynamics of higher plants in the water fluctuation zone of Yangtze River: root exudates and mass flow. Dissertation, University Düsseldorf
- Schroeder CR, Polerecky L, Klimant I (2007) Time-resolved pH/pO₂ mapping with luminescent hybrid sensors. *Anal Chem* 79:60–70
- Strobl M, Manke I, Kardjilov N, Hilger A, Dawson M, Banhart J (2009) Advances in neutron radiography and tomography. *J Phys D Appl Phys* 42:243001. doi:10.1088/0022-3727/42/24/243001
- Tournaire-Roux C, Sutka M, Javot H, Gout E, Gerbeau P, Luu DT, Bagny R, Maurel C (2003) Cytosolic pH regulates root water transport during anoxic stress through gating of aquaporins. *Nature* 425:393–397
- Tumlinson LG, Liu H, Silk WK, Hopmans JW (2008) Thermal neutron computed tomography of soil water and plant roots. *Soil Sci Soc Am J* 72:1234–1242